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Method for detecting and analyzing time-series data based on cyber threat framework

Abstract

This disclosure details a method for detecting and analyzing time-series data with a cyber threat framework. It involves determining target API events, mapping these to threat behaviors, creating threat scenarios, assessing matching degree and risk, predicting threat behaviors based on risk grades, and providing solutions.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of Disclosure

(1) The present disclosure relates to a method for detecting and analyzing time-series data based on a cyber threat framework, and more particularly, to a method for generating virtual threat scenarios based on data for API events occurring time-sequentially and analyzing and coping with a cyber threat in advance through comparison of the virtual threat scenarios.

Related Art

(2) The damage from cybersecurity threats, which are gradually becoming more sophisticated, centering on new or variant malware, has been increasing. In particular, attacks targeting vulnerabilities of servers are increasingly being carried out as zero-day attacks, which makes it easy and quick to infiltrate into the servers by exploiting vulnerabilities of web applications.

(3) In order to reduce such damage even a little and to respond at an early stage, countermeasure technology has been advancing through multi-dimensional pattern composition, various types of complex analysis, etc. However, recent cyber-attacks tend to increase day by day rather than being adequately responded to within a control range.

(4) These cyberattacks threaten finance, transportation, environment, health, etc. that directly affect lives of people beyond the existing information and communication technology (ICT) infrastructure.

(5) One of basic technologies to detect and respond to most existing cybersecurity threats is to create a database of patterns for cyberattacks or malware in advance, and utilize appropriate monitoring technologies where data flow is required.

(6) Existing technology has evolved based on a method of identifying and coping with threats when a data flow or code matching a monitored pattern is detected.

(7) Such existing technology has an advantage of being able to rapidly and accurately perform detection when a data flow or code matches a previously secured pattern. However, the technology has a problem in that, in the case of a new or mutant threat for which a pattern is not secured or is bypassed, detection is impossible or it takes a significantly long time for analysis.

(8) As a related art, Korea Patent No. 10-2419451 discloses “System And Method Of Automatizing Threat Analysis Based On Artificial Intelligence”.

(9) This related art relates to a system and a method of automatizing a threat analysis based on artificial intelligence. The system includes: a playbook automatic-generation module configured to generate a playbook based on a template by utilizing an artificial learning model; a playbook verification and management module configured to verify effectiveness of the playbook generated by the playbook automatic-generation module; a playbook database configured to save the playbook verified by the playbook verification and management module; and a playbook execution module configured to automatically execute any playbook corresponding to a detected event through matching therebetween from the playbook database. Accordingly, by automatically generating and verifying a playbook and reinforcing the playbook according to characteristics of a security control center, an environment may be provided where security control personnel (controllers) can focus only on important security events and analyze and respond to such events.

(10) However, in the aforementioned related art, since analysis and response are performed only on the generated playbook, it is not possible to respond to cyber security threats that change and evolve day by day.

Related Document

Patent Document

(11) (Patent Document 1) Korean Patent No. 10-2419451

SUMMARY OF THE DISCLOSURE

(12) One aspect of the present disclosure is to analyze a large amount of cyber security data to provide insight into cyber attack trends, techniques, and threat methods so as to effectively cope

with a cyber-attack.

(13) In one aspects, there is provided a method for detecting and analyzing time-series data based on a cyber threat framework, the method including: an event determining operating of determining, according to a predetermined criterion, at least one target API event from an event set including a plurality of events stored in a cloud environment; a mapping operation of mapping the at least one target API event to a threat behavior type corresponding to at least one technique included in a threat behavior analysis matrix based on a pre-stored threat behavior profile, wherein the threat behavior analysis matrix comprises a plurality of tactics and at least one technique included in each of the tactics; a scenario creating operation of combining at least one of the at least one target API event based on the threat behavior type and creating at least one threat scenario; a numerical value calculating operation of calculating a degree of matching and a risk for the at least one threat scenario based on a database; a risk grade determining operating of determining a risk grade of the at least one threat scenario based on at least one of the degree of matching and the risk; a threat behavior determining operating of determining a predicted threat behavior corresponding to the at least one threat scenario based on the risk grade; and a solution providing operation of creating and providing a prompt based on the predicted threat behavior.

(14) In the event determining operating, the target API event may be determined by filtering the event set according to a classification criterion comprising at least one of region, time, account, and behavior type.

(15) In the mapping operation, a threat behavior type and an individual threat behavior corresponding to the target API event may be mapped based on the threat behavior profile.

(16) The numerical value calculating operation may include calculating a degree of matching between a threat behavior type included in a past scenario pre-stored in the database and a target API event included in the at least one threat scenario by comparing the at least one threat scenario with the past scenario.

(17) The numerical value calculating operation may include a risk calculating operation of calculating, by the main server, a risk of the at least one threat scenario based on a threat behavior type included in the at least one threat scenario.

(18) In the risk grade determining operating, a risk grade of the at least one threat scenario may be determined based on an overall risk score calculated based on the degree of matching and the risk grade.

(19) In the solution providing operation, a security enhancement solution prompt corresponding to the determined predicted threat behavior may be generated and provided.

(20) In the solution providing operation, a system recovery prompt corresponding to the determined predicted threat behavior may be generated and provided.

(21) In the solution providing operation, an analysis prompt corresponding to the determined predicted threat behavior and the at least one threat scenario may be generated and provided.

(22) In the solution providing operation, an instructional prompt corresponding to the determined predicted threat behavior may be generated and provided.

(23) In the risk grade determining operating, the risk grade of the at least one threat scenario may be determined based on an overall risk score calculated by assigning weights to the degree of matching and the risk grade.

(24) The threat behavior determining operation may include: a selecting operation of selecting any one scenario from among a plurality of threat scenarios based on the risk grade; an extracting operation of extracting a past scenario having a highest degree of matching with the selected threat scenario; and a determining operation of determining a predicted threat behavior by comparing the extracted past scenario with a target API event included in the selected threat scenario.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a conceptual diagram showing a network environment according to the present disclosure.
- (2) FIG. 2 is a table showing threat behavior types and individual threat behaviors according to the present disclosure.
- (3) FIG. 3 is a flow chart showing a method for detection and analysis according to the present disclosure.
- (4) FIG. 4 is a table showing the mapping results according to the present disclosure.
- (5) FIG. 5 is a conceptual diagram illustrating an example of generating a threat scenario according to the present disclosure.
- (6) FIG. 6 is a flow chart showing an operation of calculating a numerical value according to the present disclosure.
- (7) FIG. 7 is a conceptual diagram showing an example of how to calculate a numerical value and a grade according to the present disclosure.
- (8) FIG. 8 is a conceptual diagram showing an example of how to determine a threat behavior according to the present disclosure.
- (9) FIG. 9 is a flowchart illustrating an operation of determining a threat behavior according to the present disclosure.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

- (10) Hereinafter, the present disclosure will be described in detail according to exemplary embodiments disclosed herein, with reference to the accompanying drawings. For the sake of brief description with reference to the drawings, the same or equivalent components may be provided with the same or similar reference numbers, and description thereof will not be repeated. In addition, in the following description of the embodiments, a detailed description of known functions and configurations incorporated herein will be omitted when it may impede the understanding of the embodiments.
- (11) While terms including ordinal numbers, such as “first” and “second,” etc., may be used to describe various components, such components are not limited by the above terms. The above terms are used only to distinguish one component from another.
- (12) As used herein, the singular forms “a”, “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
- (13) In this specification, operations described may be performed regardless of a listed order, except for a case where they must be performed in the listed order due to a special causal relationship.
- (14) It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including,” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
- (15) Hereinafter, the present disclosure will be described with reference to the accompanying drawings.
- (16) FIG. 1 is a conceptual diagram showing a network environment according to the present disclosure, and FIG. 2 is a table showing threat behavior types and individual threat behaviors according to the present disclosure.
- (17) Referring to FIGS. 1 and 2, a system for detecting and analyzing time-series data based on a cyber threat framework according to the present disclosure may include a cloud server **10**, a database **20**, a user terminal **30**, an AI engine **40**, and a main server **50**.
- (18) In this case, the cloud server **10**, the database **20**, the user terminal **30**, the AI engine **40**, and the main server **50** may be connected to each other based on communication, that is, a network. Here, the network is not limited in communication scheme, and the communication scheme may

include not only a communication scheme to utilize a telecommunication network (for example, a mobile communication network, wired Internet, wireless Internet, and a broadcast network), but also a communication scheme to utilize an electric signal (for example, an analog signal, a digital signal, a signal sigil, and Pulse Width Modulation) and a short-range radio communication scheme.

(19) The cloud server **10** refers to a server for a cloud service which is a service where an external or outsourced server and a storage are used instead of a server and a storage provided by a company. Here, the cloud server **10** may include a public data center (such as Amazon's AWS and Microsoft's Azure) for public clouds and on-premises data centers for private clouds.

(20) The cloud server **10** may be implemented as a single computer device or a plurality of computer devices providing commands, codes, files, contents, services, and the like. The cloud server **10** may transmit and receive information in communication with other servers and terminals in the system through a network.

(21) The cloud server **10** may store every data and infrastructure of users utilizing the cloud server **10**. In this case, a specific user or user group may use a single cloud server **10** or may use a plurality of cloud servers **10**.

(22) Preferably, when the plurality of cloud servers **10** is used, an integrated control server for managing data distributed and stored in the plurality of cloud servers **10** may be provided separately. More preferably, the main server **50** to be described later may serve as the integrated control server.

(23) The user terminal **30** is an entity that accesses a specific company or organization's data stored in the cloud server **10** to read and write data. In this case, the user terminal **30** with authorized access to the cloud server **10** may be a member of a company, organization, or group. If an unauthorized user attempts to access the cloud server **10** of the company, organization, or group, this access may be regarded as a cyber threat or attack, which is commonly known as hacking.

(24) The user terminal **30** may be implemented as a computer device or a plurality of computer devices providing commands, codes, files, contents, services, and the like. The cloud server **10** may transmit and receive information in communication with other servers and terminals in the system through a network.

(25) The database **20** may serve as a storage medium for storing data. Preferably, in the present disclosure, the database **20** may store cyber threats occurred in the past, that is, scenarios of hacking.

(26) The database **20** may be implemented as a computer device or a plurality of computer devices providing commands, codes, files, contents, services, and the like. In addition, the database **20** may transmit and receive information in communication with other servers and terminals in the system through a network.

(27) Furthermore, the database **20** may store information on actions (defensive measures) that defenders (administrators) can take to prevent and detect cyber-attacks, and may also store analyzed data on named hacking groups and their attack techniques.

(28) Here, a scenario refers to a process in which an attacker performs step-by-step actions to carry out a cyber-attack. In such a scenario, actions performed by the attacker at each step are listed time-sequentially.

(29) Therefore, the database **20** may basically store scenarios occurring the past, that is, past scenarios. Preferably, in this case, a scenario may be a kill chain stored in MITRE ATT&CK (Adversarial Tactics, Techniques and Common Knowledge). Specifically, MITRE ATT&CK is a cyber security-related knowledge base created by MITRE Corporation, and it is composed of tactics with categories for cyber attackers' goals and steps, and techniques which are specific methods for the tactics.

(30) Accordingly, a threat behavior is included in at least one tactic and corresponds to at least one technique included in the tactic.

(31) In addition, the scenarios stored in this database **20** are data that is classified and listed

information on various attack groups' techniques, which are analyzed in terms of tactics and techniques, regarding adversary behaviors employed by attackers in cyber-attacks. Specifically, a scenario includes at least one technique, which is a single step, as a method for an attacker to achieve at least one tactic, which is an attack goal. For example, a threat behavior type corresponding to a technique may include Spearphishing via Email, Drive-by Compromise, Credential Dumping, Man-in-the-Middle attack (MitM), Command and Control over alternative protocol, etc.

(32) The AI engine **40** may refer to an existing artificial intelligence (AI) engine. That is, the AI engine **40** may refer to any of various existing AI engines **40**. Such an AI engine **40** may provide a data filtering or mapping function.

(33) The AI engine **40** may be implemented as a computer device or a plurality of computer devices providing commands, codes, files, contents, services, and the like. In addition, the AI engine **40** may transmit and receive information in communication with other servers and terminals in the system through a network.

(34) As described above, the main server **50** performs the integrated control function for the plurality of cloud servers **10** and, at the same time, detects and analyzes cyber threats to provide various prompts to an authorized user terminal **30**, that is, a user terminal of a defender (administrator).

(35) The main server **50** may be implemented as a computer device or a plurality of computer devices providing commands, codes, files, contents, services, and the like. The main server **50** may transmit and receive information in communication with other servers and terminals in the system through a network.

(36) FIG. **3** is a flow chart showing a method for detection and analysis according to the present disclosure; FIG. **4** is a table showing the mapping results according to the present disclosure; FIG. **5** is a conceptual diagram illustrating an example of generating a threat scenario according to the present disclosure; FIG. **6** is a flow chart showing an operation of calculating a numerical value according to the present disclosure; FIG. **7** is a conceptual diagram showing an example of how to calculate a numerical value and a grade according to the present disclosure; FIG. **8** is a conceptual diagram showing an example of how to determine a threat behavior according to the present disclosure; and FIG. **9** is a flowchart illustrating an operation of determining a threat behavior according to the present disclosure.

(37) Referring to FIGS. **3** to **9**, a method for detecting and analyzing time-series data based on a cyber threat framework according to the present disclosure may include an event determining operation **S10**, a mapping operation **S20**, a scenario creating operation **S30**, a numerical value calculating operation **S40**, a risk grade determining operation **S50**, a threat behavior determining operation **S60**, and a solution providing operation **S70**.

(38) In the event determining operation **S10**, according to a predetermined criterion, the main server **50** may determine at least one target API event from an event set including a plurality of events stored in a cloud environment established through the cloud server **10**.

(39) In the cloud server **10**, every action (event) occurring due to the user terminal **30** in a web page or application of a corresponding company, organization, or group is stored in the form of an API. In this case, an event set including a plurality of consecutive events occurring due to the user terminal **30** is generated.

(40) Accordingly, the event set including the plurality of events may be stored in the cloud environment. Here, according to a predetermined criterion, a target API event to be detected and analyzed may be determined from among the events included in the stored event set.

(41) In this case, filtering may be performed on the event set according to the predetermined criterion to determine a plurality of target API events.

(42) In one embodiment, the target API event may be determined by performing filtering on a plurality of events according to a classification condition including at least one of region, time,

account, and behavior type.

(43) Here, the region may be referred to as a region where an API is created, and the region may be identified based on an IP where the API is created. Therefore, by performing filtering on the stored event set by region of creation, only an event occurring in a specific region may be determined as a target API event.

(44) The time may refer to a time when an API is created, and only an event occurring in a specific time period may be determined as a target API event by filtering the stored event set according to a created time period.

(45) The account may be referred to as an account in which an API is created, and by filtering only a specific account from accounts which have accessed a corresponding website or application and performed an action, only an event occurring in the specific account in the event set may be determined as a target API event.

(46) In this case, a risk of the threat scenario may be a maximum value of risks of a plurality of threat behavior types included in the threat scenario.

(47) In one embodiment, the numerical value calculating operation **S40** may include a degree-of-matching calculating operation **S41** and a risk calculating operation **S42**.

(48) In the degree-of-matching calculating operation **S41**, the main server **50** compares the threat scenario with a past scenario pre-stored in the database **20** to calculate a degree of matching between a threat behavior type included in the past scenario and a target API event included in the threat scenario.

(49) The past scenario stored in the database **20** refer to a scenario of attacks performed by an attacker in the past, that is, a scenario of threat behaviors. In other words, threat behavior types actually performed by an attacker in the past and individual threat behaviors included in the threat behavior types are listed time-sequentially in the past scenario.

(50) Therefore, a past scenario include a plurality of threat behavior types, and a threat scenario also include at least one target API event. Therefore, based on calculated degrees of matching between each past scenario and each threat scenario, a threat scenario having a highest degree of matching with the past scenarios stored in the database **20** may be identified among threat scenarios.

(51) In this case, a degree of matching may be shown as a numerical value, and a degree of similarity between a past scenario and a threat scenario may be calculated as the degree of matching.

(52) Here, preferably, the degrees of matching with all past scenarios stored in the database **20** and a corresponding threat scenario may be calculated, and a maximum value of the calculated degrees of matching may be determined as the degree of matching for the threat scenario.

(53) In the risk calculating operation **S42**, the main server **50** may calculate a risk of a threat scenario according to a threat behavior type included in the threat scenario.

(54) Here, the risk of the threat scenario may be a maximum value of risks of the threat behavior types included in the threat scenario.

(55) Therefore, preferably, when calculating a risk, a process of presetting a risk of every threat behavior type included in the database **20** may be performed.

(56) Thereafter, a threat behavior type included in the threat scenario may be compared with the database **20** to calculate a risk of every threat behavior type included in the threat scenario and determine a maximum value of the calculated risks as a risk of the threat scenario.

(57) For example, if a threat scenario includes only a threat behavior type of initial access, the risk is relatively low, but if a threat scenario includes credential access, the risk is higher than that of the threat scenario including only initial access.

(58) In the risk grade determining operation **S50**, the main server **50** may determine a risk grade of the threat scenario based on at least one of the degree of matching and the risk.

(59) The risk grade may be determined based on how much the threat scenario matches a past

scenario (degree of matching) and how high the risk of the threat activity included in the threat scenario is (risk).

(60) Here, a method of determining a risk grade based on a degree of matching and a risk calculated for a threat scenario is not limited. Preferably, if at least one of the degree of matching and the risk is higher or if the degree of matching and the risk are higher, the higher risk grade may be set.

(61) In this case, in the risk grade determining operation **S50**, a risk grade of the threat scenario may be determined based on at least one of the degree of matching and the risk. More preferably, the risk grade of the threat scenario may be determined based on the degree of matching and the risk. In other words, the risk grade of the threat scenario may be determined in consideration of both the degree of matching and the risk.

(62) In one embodiment, an overall risk score may be calculated based on the degree of matching and the risk, and a risk grade of the threat scenario may be determined based on the overall risk score.

(63) In this case, in calculating the overall risk score, the same weights may be assigned to the degree of matching and the risk. In one embodiment, however, the overall risk score may be calculated by assigning different weights to the degree of matching and the risk.

(64) In other words, the degree of matching and the risk grade may have different weights. The weights respectively assigned to the degree of matching and the risk grade may be values set by an administrator (defender). There is no separate limitation in the weights.

(65) Therefore, a risk grade of each threat scenario may be set based on an overall risk score calculated by assigning weights to a degree of matching and a risk.

(66) In the threat behavior determining operation **S60**, the main server **50** determines a threat behavior corresponding to the threat scenario based on the risk grade.

(67) Here, determining a threat behavior in the threat behavior determining operation **S60** may be determining a threat behavior (target API event) belonging to a threat scenario having a highest risk grade and a predicted threat behavior to be additionally included in the threat scenario.

(68) In other words, the threat behavior determining operation **S60** may be an operation of determining a predicted threat behavior, which is an individual threat behavior not included in the threat scenario but predicted to be additionally included in the threat scenario based on past scenarios.

(69) In one embodiment, the threat behavior determining operation **S61** may include a selecting operation **S61**, an extracting operation **S62**, and a determining operation **S63**.

(70) In the selecting operation **S61**, the main server **50** selects one of a plurality of threat scenarios based on the risk grade. Preferably, a threat scenario having a highest risk grade may be selected, or only a plurality of threat scenarios having high risk grades may be selected in descending order of risk grades.

(71) In the extracting operation **S62**, the main server **50** extracts a past scenario having a highest degree of matching with the selected threat scenario. In the above-described degree-of-matching calculating operation **S41**, it has been described that the degree of matching between a threat scenario and each past scenario is calculated and a maximum value of calculated consistencies is the degree of matching for the threat scenario. Thus, the past scenario determined to have a highest degree of matching with the corresponding threat scenario is extracted.

(72) In the determining operation **S63**, the main server **50** compares the extracted past scenario with the threat behavior type included in the selected threat scenario to determine a predicted threat behavior.

(73) Here, the predicted threat behavior refers to an individual threat behavior that is included in the threat behavior type from the extracted past scenario, but not included in the current threat scenario.

(74) In other words, in the extracted historical scenario, there may be a threat behavior type that is

not included in the threat scenarios, and such a threat behavior type include individual threat behaviors. Thus, any individual threat behavior predicted to be performed by an attacker according to the threat scenario is determined as a predicted threat behavior.

(75) In the solution providing operation S70, the main server 50 may generate a prompt based on the determined predicted threat behavior and provide the user terminal 30 with the prompt.

(76) For example, in the solution providing operation S70, the main server 50 may generate a security enhancement solution prompt corresponding to the determined predicted threat behavior and provide the user terminal 30 with the security enhancement solution prompt.

(77) Here, the security enhancement solution prompt may be a prompt associated with a solution that compensates for vulnerabilities of the determined predicted threat behavior and enhances security settings. The security enhancement solution prompt may include security setting guides, patch and update guides, and optimal setting methods for security tools.

(78) The security enhancement solution prompt may also provide intelligence on how to detect and cope with a determined predicted threat behavior. Here, the security enhancement solution prompt may include a method for improving a threat detecting algorithm, a method for applying anomaly detection, and standard operating procedures (SOPs) for quick response.

(79) For example, in the solution providing operation S70, the main server 50 may generate a system recovery prompt corresponding to the determined predicted threat behavior and provide the user terminal 30 with the system recovery prompt.

(80) Here, the system recovery prompt may include a guide for system recovery and best practices for a recovery process. Furthermore, the system recovery prompt may include guides on how to minimize an additional threat caused by the determined predicted threat behavior, and instructions required to restore the system to a safe state.

(81) For example, in the solution providing operation S70, the main server 50 may generate an analysis prompt corresponding to the determined predicted threat behavior and the threat scenario and provide the user terminal 30 with the analysis prompt.

(82) Here, the analysis prompt may provide detailed log information on each individual API event involved in the threat scenario, providing information on which TTPS was used, which user or system was targeted, which data was at risk, and which parts of the system were vulnerable.

(83) For example, in the solution providing operation S70, the main server 50 may generate an instructional prompt corresponding to the determined predicted threat behavior and provide the user terminal 30 with the instructional prompt.

(84) Here, the instructional prompt may be a message or material for training a user in preparation for the determined predicted threat behavior. The instructional prompt may include a description of the determined predicted threat behavior, precautions, safe use methods, and the like.

(85) In the method for detecting and analyzing time-series data based on a cyber threat framework of the present invention, it is possible to analyze a large amount of cyber security data to provide insight into cyber attack trends, techniques, and threat methods so as to effectively cope with a cyber-attack.

(86) The technical features disclosed in each embodiment of the present disclosure are not limited to a corresponding embodiment, and unless incompatible with each other, the technical features disclosed in each embodiment may be applied in combination to other embodiments.

(87) Although embodiments of the present disclosure are described mainly about individual technical features, the technical features of the embodiments may be applied in combination, unless incompatible with each other.

(88) The present disclosure is not limited to the above-described embodiments and the accompanying drawings, and various modifications and changes may be made in view of a person skilled in the art to which the present disclosure pertains. Therefore, the scope of the present disclosure should be defined by not only the claims of this specification but also those equivalent to these claims.

Claims

1. A method for detecting and analyzing time-series data based on a cyber threat framework, the method comprising: an event determining operation of determining, according to a predetermined criterion, at least one target API event from an event set including a plurality of events stored in a cloud environment; a mapping operation of mapping the at least one target API event to a threat behavior type corresponding to at least one technique included in a threat behavior analysis matrix based on a pre-stored threat behavior profile, wherein the threat behavior analysis matrix comprises a plurality of tactics and at least one technique included in each of the tactics; a scenario creating operation of combining at least one of the at least one target API event based on the threat behavior type and creating at least one threat scenario; a numerical value calculating operation of calculating a degree of matching and a risk for the at least one threat scenario based on a database, wherein the numerical value calculating operation comprises a risk calculating operation of calculating, by the main server, a risk of the at least one threat scenario based on a threat behavior type included in the at least one threat scenario; a risk grade determining operation of determining a risk grade of the at least one threat scenario based on at least one of the degree of matching and the risk, wherein in the risk grade determining operation, a risk grade of the at least one threat scenario is determined based on an overall risk score calculated based on the degree of matching and the risk; a threat behavior determining operation of determining a predicted threat behavior corresponding to the at least one threat scenario based on the risk grade; and a solution providing operation of creating and providing a prompt based on the predicted threat behavior.
2. The method of claim 1, wherein in the event determining operation, the target API event is determined by filtering the event set according to a classification criterion comprising at least one of region, time, account, and behavior type.
3. The method of claim 1, wherein in the mapping operation, a threat behavior type and an individual threat behavior corresponding to the target API event are mapped based on the threat behavior profile.
4. The method of claim 1, wherein the numerical value calculating operation comprises calculating a degree of matching between a threat behavior type included in a past scenario pre-stored in the database and a target API event included in the at least one threat scenario by comparing the at least one threat scenario with the past scenario.
5. The method of claim 1, wherein in the solution providing operation, a security enhancement solution prompt corresponding to the determined predicted threat behavior is generated and provided.
6. The method of claim 1, wherein in the solution providing operation, a system recovery prompt corresponding to the determined predicted threat behavior is generated and provided.
7. The method of claim 1, wherein in the solution providing operation, an analysis prompt corresponding to the determined predicted threat behavior and the at least one threat scenario is generated and provided.
8. The method of claim 1, wherein in the solution providing operation, an instructional prompt corresponding to the determined predicted threat behavior is generated and provided.
9. The method of claim 1, wherein, in the risk grade determining operation, the risk grade of the at least one threat scenario is determined based on an overall risk score calculated by assigning weights to the degree of matching and the risk grade.
10. The method of claim 1, wherein the threat behavior determining operation comprises: a selecting operation of selecting any one threat scenario from among a plurality of threat scenarios based on the risk grade; an extracting operation of extracting a past scenario having a highest degree of matching with the selected threat scenario; and a determining operation of determining a

predicted threat behavior by comparing the extracted past scenario with a target API event included in the selected threat scenario.
